

Informational Holonomy and the Electron g-Factor

A Conditional Selection Theorem, Exact Dirac Reduction, and an Environmental SMEFT Bridge

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Abstract

Place an electron in a magnetic field and its spin-dependent magnetic moment precesses. Any proposed microscopic account of electromagnetism must therefore explain not only that a phase can be transported, but why the low-energy electron couples with the standard Dirac magnetic coefficient and where a measurable departure could enter. This paper separates that problem into four layers. First, established gauge geometry represents electromagnetism by a $U(1)$ connection and its holonomy. Second, a bounded operator-selection theorem shows that, after the matter representation and a precise phase-only-entry design space are fixed, the renormalizable bilinear matter sector reduces to the minimally coupled Dirac form. Holonomy alone does not select a Dirac field; the theorem is conditional on the declared design space. Third, an explicit Foldy-Wouthuysen reduction and the complementary Gordon identity recover the standard tree-level value $g = 2$, while quantum electrodynamics and higher-dimensional operators supply separate corrections. Fourth, a gauge-invariant environmental extension promotes the electroweak dipole Wilson coefficients to functions of dimensionless Lorentz scalars constructed from proposed Super Information Theory (SIT) time-density and coherence fields. This produces a differential, falsifiable response family without claiming that its coefficients have already been calculated or observed. The paper also distinguishes electromagnetic $U(1)$ holonomy from spinor 4π periodicity, restores the precession-sign account, and states the measurement and failure conditions needed to turn the environmental bridge into a quantitative SIT prediction.

equation; Foldy-Wouthuysen transformation; Gordon identity; SMEFT; Super Information Theory; environmental Wilson coefficient

Keywords: electron magnetic moment; gyromagnetic ratio; holonomy; Dirac

1. The physical problem before the new vocabulary

An electron in a magnetic field supplies an unusually sharp test of a physical theory. The electron has spin angular momentum, the spin is associated with a magnetic moment, and the moment changes the energy and precession of the electron in a field. For a particle of mass m , charge q , spin S , and magnetic moment μ , define the gyromagnetic factor by

$$\mu = g \frac{q}{2m} S. \quad (1)$$

For the electron, $q = -e$, where $e > 0$ is the elementary-charge magnitude. The magnetic moment therefore points opposite to the spin when g is positive under this convention.

The unexplained problem is not merely why an electromagnetic phase exists. A theory must account for a chain:

1. a charged spinor changes phase under local gauge transport;
2. the phase relation becomes a covariant derivative in the action;

3. the relativistic action has a definite nonrelativistic spin-field term;
4. that term produces the tree-level coefficient $g = 2$;
5. loops and higher-dimensional interactions can modify the observable.

Established Dirac theory already supplies steps 2-4, and quantum electrodynamics supplies the leading radiative corrections. The question for SIT is narrower and more useful: can its proposed informational phase geometry recover the established chain under explicit assumptions, and can it specify where a genuinely SIT-dependent response would enter?

The measured quantity is often expressed through the anomaly

$$a_e \equiv \frac{g_e - 2}{2}, \quad \frac{g_e}{2} = 1 + a_e. \quad (2)$$

At tree level in minimal Dirac theory, $a_e = 0$. The physical electron has a nonzero anomaly. Keeping those two statements separate prevents a standard tree result from being mistaken for a complete prediction.

2. Established geometry: connection, curvature, and holonomy

Let $P \rightarrow M$ be a principal $U(1)$ bundle over spacetime. In a local section, the electromagnetic connection and curvature are represented by

$$A = A_\mu dx^\mu, \quad F = dA. \quad (3)$$

In natural units, with the convention used throughout this paper,

$$D_\mu = \partial_\mu + iqA_\mu, \quad \psi \mapsto e^{-iq\alpha}\psi, \quad A_\mu \mapsto A_\mu + \partial_\mu\alpha. \quad (4)$$

Parallel transport along a path γ produces the Wilson-line factor

$$U[\gamma] = \exp\left(iq \int_\gamma A\right). \quad (5)$$

This is the established scientific account that SIT inherits. A connection is not generally a gradient. On a pure-gauge patch one can write $A = d\theta$ up to conventions, but a general connection is not exact. The Aharonov-Bohm effect further shows why local curvature along accessible paths and global holonomy must not be conflated.

SIT proposes an interpretation of this structure: the $U(1)$ phase fiber is a coherence-phase channel, so electromagnetic parallel transport is read as an informational phase relation. That interpretation is not used as a substitute for the bundle mathematics. It becomes physically consequential only if a larger SIT model derives the connection, its action, or a distinctive coefficient relation.

3. The phase-only-entry design problem

Holonomy tells us how a chosen charged representation is transported. It does not tell us whether matter is a scalar, spinor, vector, or composite. It also does not choose derivative order, parity, an operator cutoff, or the absence of additional fields. The earlier version of this paper compressed those choices into the word “unique.” The correction is to expose the design space, not to discard the valid conditional result.

3.1 Frozen design space

Consider a matter sector satisfying all of the following:

1. The dynamical matter field is one fixed four-component Dirac spinor ψ of mass m and charge q .
2. Operators are local, Lorentz scalar, Hermitian, bilinear in ψ , and contain at most one derivative.
3. Canonical mass dimension is at most four.
4. The retained matter sector is parity even and contains no independent background tensor or additional matter field.
5. The electromagnetic connection enters this sector only through the covariant derivative D_μ . This is the **phase-only-entry condition**.
6. Overall kinetic normalization, the real mass parameter, and total derivatives are not counted as distinct operator structures.

These premises are strong, visible, and testable as a design contract. They are not consequences of holonomy alone.

3.2 Conditional selection theorem

Theorem 1 - Phase-only-entry selection. Within the frozen design space above, the matter Lagrangian is, up to the stated equivalences,

$$\mathcal{L}_D = \bar{\psi} (i\gamma^\mu D_\mu - m) \psi. \quad (6)$$

Proof sketch. A Dirac spinor has canonical dimension $3/2$. A bilinear without derivatives therefore has dimension three. Parity evenness retains the scalar mass bilinear $\bar{\psi}\psi$ and excludes the pseudoscalar mass bilinear from this design space. A bilinear with one derivative has dimension four. Lorentz invariance, parity evenness, and the absence of external tensors select the vector contraction $\bar{\psi}\gamma^\mu D_\mu\psi$; integration by parts identifies the corresponding left-acting derivative form, and Hermiticity fixes their combination. Local U(1) covariance fixes the connection term inside D_μ relative to the derivative. An explicit field-strength dipole $\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}$ has dimension five and is outside the cutoff. This leaves Equation (6), its normalization, its mass, and total derivatives. QED

The theorem is a bounded operator-classification statement:

$$\text{fixed Dirac design space} + \text{phase-only entry} \implies \mathcal{L}_D. \quad (7)$$

It is not the invalid statement holonomy $\rightarrow$ Dirac field. The contribution is to identify the assumptions under which the informational phase interpretation selects the minimal charged-spinor operator rather than allowing an independent tree-level dipole in the renormalizable matter sector.

4. Exact tree-level reduction

The value $g = 2$ is a standard Dirac result. Reproducing the coefficient explicitly matters because it reveals exactly where the factor enters and which assumptions would need to change to alter it.

4.1 Even and odd operators

For static external fields, write the minimally coupled Dirac Hamiltonian as

$$\begin{aligned}
H &= \beta m + \mathcal{E} + \mathcal{O}, \\
\mathcal{E} &= q\phi, \\
\mathcal{O} &= \boldsymbol{\alpha} \cdot \boldsymbol{\Pi}, \\
\boldsymbol{\Pi} &= \mathbf{p} - q\mathbf{A},
\end{aligned} \tag{8}$$

where $[\beta, \mathcal{E}] = 0$ and $\{\beta, \mathcal{O}\} = 0$. The Foldy-Wouthuysen expansion through the terms needed here is

$$H_{\text{FW}} = \beta \left(m + \frac{\mathcal{O}^2}{2m} \right) + \mathcal{E} - \frac{1}{8m^2} [\mathcal{O}, [\mathcal{O}, \mathcal{E}]] + O(m^{-3}). \tag{9}$$

The magnetic coefficient comes from $\mathcal{O}^2$. Under the convention

$$[\Pi_i, \Pi_j] = iq \epsilon_{ijk} B_k, \quad \alpha_i \alpha_j = \delta_{ij} + i \epsilon_{ijk} \Sigma_k, \tag{10}$$

the antisymmetric contraction gives

$$\begin{aligned}
(\boldsymbol{\alpha} \cdot \boldsymbol{\Pi})^2 &= \Pi^2 + \frac{i}{2} \epsilon_{ijk} \Sigma_k [\Pi_i, \Pi_j] \\
&= \Pi^2 - q \boldsymbol{\Sigma} \cdot \mathbf{B},
\end{aligned} \tag{11}$$

because $\sum_{i,j} \epsilon_{ijk} \epsilon_{ijl} = 2\delta_{kl}$.

4.2 Positive-energy sector

Setting $\beta = +1$ for the positive-energy block yields

$$H_{\text{FW}}^{(+)} = m + q\phi + \frac{\Pi^2}{2m} - \frac{q}{2m} \boldsymbol{\sigma} \cdot \mathbf{B} + \text{Darwin and spin-orbit terms} + O(m^{-3}). \tag{12}$$

Since $S = \sigma/2$ in natural units, the spin interaction is

$$H_{\text{spin}} = -\frac{q}{m} \mathbf{S} \cdot \mathbf{B} = -g \frac{q}{2m} \mathbf{S} \cdot \mathbf{B}, \quad g_{\text{tree}} = 2. \tag{13}$$

No additional informational coefficient appears in this reduction. The conditional SIT result is therefore: if the phase-only-entry theorem governs the low-energy matter sector, the standard Dirac coefficient follows.

5. The complementary Gordon route

The same tree-level coefficient appears in the current matrix element. For on-shell free spinors and momentum transfer $k = p' - p$, the Gordon identity is

$$\bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') \left[\frac{(p' + p)^\mu}{2m} + \frac{i \sigma^{\mu\nu} k_\nu}{2m} \right] u(p). \tag{14}$$

The general on-shell electromagnetic vertex is conventionally written

$$\Gamma^\mu(k) = F_1(k^2)\gamma^\mu + \frac{i\sigma^{\mu\nu}k_\nu}{2m}F_2(k^2). \quad (15)$$

Charge normalization gives $F_1(0) = 1$. At tree level in minimal Dirac theory, $F_2(0) = 0$; therefore

$$a_e = F_2(0), \quad g_e = 2[F_1(0) + F_2(0)]. \quad (16)$$

The Foldy-Wouthuysen and Gordon routes are complementary. The first exposes the nonrelativistic Hamiltonian; the second identifies the Dirac and Pauli form factors. Their agreement is a useful internal consistency check.

6. Spinor 4-pi periodicity is a related but distinct layer

A spin-one-half state transforms under spatial rotation as

$$U(\theta) = \exp(-i\theta \hat{\mathbf{n}} \cdot \mathbf{S}), \quad U(2\pi) = -I, \quad U(4\pi) = I. \quad (17)$$

This is the double-cover relationship between $SU(2)$ and $SO(3)$. It explains why a spinor needs a 4-pi rotation to return to the same state vector. It is not the same mathematical object as electromagnetic $U(1)$ holonomy in Equation (5). The two structures can be placed in one informational narrative because both track physically meaningful phase under transport, but neither is derived from the other in this paper.

That distinction strengthens the SIT interpretation. It avoids claiming that the electron's factor of two is numerology produced by the phrase “4-pi.” The coefficient follows from the minimally coupled spinor operator and its nonrelativistic reduction. Spinor periodicity supplies representation-theory context for why spinorial phase transport differs from classical vector rotation.

7. Precession and sign conventions

Define the signed gyromagnetic ratio $\gamma = gq/(2m)$. The torque equation can be written

$$\begin{aligned} \dot{\mathbf{S}} &= \boldsymbol{\mu} \times \mathbf{B} = \gamma \mathbf{S} \times \mathbf{B} = \boldsymbol{\Omega} \times \mathbf{S}, \\ \boldsymbol{\Omega} &= -\gamma \mathbf{B}. \end{aligned} \quad (18)$$

For an electron, $q = -e$, so $\gamma < 0$ when $g > 0$. This determines the signed precession vector under the declared convention. Verbal labels such as “clockwise” depend on the viewing direction; Equation (18) is the invariant statement that should control diagrams and experimental comparisons.

8. Tree value, quantum anomaly, and measurement

Schwinger's leading one-loop quantum-electrodynamic correction is

$$a_e^{(1)} = \frac{\alpha}{2\pi}. \quad (19)$$

This is only the first term of a perturbative expansion. Precision Standard Model calculations require higher-order QED terms and smaller hadronic and electroweak contributions. They also depend on an independently determined fine-structure constant.

Fan and collaborators reported an electron magnetic-moment ratio corresponding to

$$\frac{g_e}{2} = 1.00115965218059(13) \quad (20)$$

under their experimental convention. The directly converted anomaly is $a_e = 0.00115965218059(13)$. Subtracting only Equation (19) would not produce a valid new-physics residual. Using an α value inferred from the same electron anomaly would also be circular. For these reasons, this paper reports no experimental SIT bound.

9. Where a nonminimal response can enter

Below electroweak symmetry breaking, a CP-even photon dipole can be normalized as

$$\Delta\mathcal{L}_P = \frac{e\Delta a_e}{4m_e} \bar{e}\sigma^{\mu\nu}e F_{\mu\nu}. \quad (21)$$

The operator has mass dimension five. It is absent from the dimension-four design space of Theorem 1 but can arise through loops or higher-dimensional physics.

Above electroweak symmetry breaking, the gauge-invariant Warsaw-basis operators are

$$\begin{aligned} Q_{eB} &= (\bar{l}\sigma^{\mu\nu}e)\varphi B_{\mu\nu}, \\ Q_{eW} &= (\bar{l}\sigma^{\mu\nu}e)\tau^I\varphi W_{\mu\nu}^I, \\ \Delta\mathcal{L}_6 &= \frac{C_{eB}Q_{eB} + C_{eW}Q_{eW}}{\Lambda^2} + \text{h.c.} \end{aligned} \quad (22)$$

for the first-generation flavor component. With Warsaw τ^I equal to the Pauli matrices and $\varphi \rightarrow (0, v/\sqrt{2})^T$, define

$$C_{e\gamma} = c_W C_{eB} - s_W C_{eW}. \quad (23)$$

The CP-even photon coefficient is $v \operatorname{Re}(C_{e\gamma})/(\sqrt{2}\Lambda^2)$. Matching Equation (22) to Equation (21) gives

$$\Delta a_e = \frac{2\sqrt{2}m_e v}{e\Lambda^2} \operatorname{Re} C_{e\gamma}. \quad (24)$$

Equation (24) is a tree-level coefficient dictionary. A precision phenomenological analysis must include running, mixing, matching conventions, input-scheme choices, and other operator contributions. Those requirements are why the Aebischer et al. one-loop SMEFT-to-LEFT treatment cannot be replaced by a single scale estimate.

10. The SIT environmental response family

The earlier paper proposed that time-density and coherence gradients could modulate a dipole coefficient. The problem was not the idea of an environment-dependent coefficient. The problem was that the old expression did not define a gauge-invariant electroweak operator, close the dimensions, or specify the measured contrast.

Assume, conditionally, that SIT supplies scalar singlet fields ρ_t and R_{coh} . Let d_ρ be the mass dimension of ρ_t , and take R_{coh} to be dimensionless for illustration. Dimensionless Lorentz scalars include

$$\begin{aligned} X_\rho &= \frac{\partial_\mu \rho_t \partial^\mu \rho_t}{M_\rho^{2d_\rho+2}}, \\ X_R &= \frac{\partial_\mu R_{\text{coh}} \partial^\mu R_{\text{coh}}}{M_R^2}, \\ X_{\rho R} &= \frac{\partial_\mu \rho_t \partial^\mu R_{\text{coh}}}{M_{\rho R}^{d_\rho+2}}. \end{aligned} \tag{25}$$

Other invariant choices are possible. Equation (25) is valuable because it makes the transformation and dimensional assumptions explicit. A laboratory rest-frame derivative such as $u^\mu \partial_\mu \rho_t$ would instead introduce a preferred timelike vector and must be treated as a Lorentz-breaking extension, not silently inserted into ordinary SMEFT.

Promote the Wilson coefficients to response functions,

$$\begin{aligned} C_{eB}(\mu, X) &= C_{eB}^{(0)}(\mu) + \kappa_{eB}^\rho X_\rho + \kappa_{eB}^R X_R + \kappa_{eB}^{\rho R} X_{\rho R} + \dots, \\ C_{eW}(\mu, X) &= C_{eW}^{(0)}(\mu) + \kappa_{eW}^\rho X_\rho + \kappa_{eW}^R X_R + \kappa_{eW}^{\rho R} X_{\rho R} + \dots. \end{aligned} \tag{26}$$

The corresponding photon response is $\kappa_{e\gamma}^a = c_W \kappa_{eB}^a - s_W \kappa_{eW}^a$. Between two matched environmental states X_A and X_B , the leading differential anomaly is

$$\Delta a_e(X_A) - \Delta a_e(X_B) = \frac{2\sqrt{2} m_e v}{e\Lambda^2} \text{Re} \left[\sum_a \kappa_{e\gamma}^a (X_A^a - X_B^a) \right] + \dots \tag{27}$$

This equation recovers the intended zero-gradient limit without pretending to calculate an effect:

$$X_\rho = X_R = X_{\rho R} = 0 \implies \Delta a_e^{\text{environmental}} = 0 \tag{28}$$

for the environmental component of this parameterization.

Equations (25)-(28) are the new proposed operation. The convenient name **environmental SMEFT bridge** is introduced only after its function is clear: proposed SIT state variables are translated into dimensionless invariants, those invariants modulate gauge-invariant Wilson coefficients, and the coefficients map to a differential magnetic-moment observable.

11. Discriminating experiment and failure conditions

A scientific test must compare more than a metaphorical “coherent” and “decoherent” setting. It requires:

1. operational definitions and units for ρ_t and R_{coh} ;
2. a reproducible intervention that changes at least one X in Equation (25);
3. matched control of magnetic field, electric field, trap geometry, temperature, motion, charge state, calibration drift, and ordinary material systematics;
4. a preregistered response coefficient or bounded parameter range;
5. an independently chosen α and complete Standard Model prediction if an absolute anomaly residual is used;
6. repeated reversal or modulation of the proposed environmental state;
7. blinded analysis of the differential frequency ratio.

The parameterized submodel fails if a controlled intervention produces a well-resolved $X_A - X_B$ but the observed differential excludes its preregistered $\kappa_{e\gamma}$ range. A response that tracks temperature, field drift, apparatus geometry, or another conventional variable rather than the defined X also fails the proposed interpretation. Conversely, a correlation is not sufficient: it must reproduce under reversal, survive systematic controls, and obey the coefficient and scale dependence of Equation (27).

An unspecified coefficient cannot make a numerical prediction. Therefore a null experiment against the present family does not falsify every possible SIT model. It constrains or excludes only the operationally completed submodel. This is a feature of honest hypothesis construction, not a retreat from falsifiability.

12. Claim atlas and genealogy

Claim ID	Claim	Status	Established source or author layer	Decisive test
EGF-01	Electromagnetism is represented locally by a $U(1)$ connection	Established	Gauge theory; Dirac and Aharonov-Bohm lineage	Standard gauge consistency
EGF-02	SIT interprets the connection as informational coherence-phase transport	SIT interpretation	Version 6, retained here	Derive substrate dynamics or a unique coefficient relation
EGF-03	The frozen phase-only-entry design space selects the minimal Dirac bilinear	Conditional author theorem	Theorem 1	Exhibit an allowed counteroperator satisfying every premise
EGF-04	Minimal Dirac reduction gives tree-level $g = 2$	Established result reproduced	Dirac; Foldy-Wouthuysen	Algebraic reduction
EGF-05	Gordon and Foldy-Wouthuysen routes agree	Established result reproduced	Standard QFT/QM	Form-factor and Hamiltonian coefficient match
EGF-06	Spinor 4-pi periodicity and $U(1)$ holonomy are distinct phase structures	Established distinction plus SIT interpretation	Spin representation and gauge geometry	Representation-level derivation
EGF-07	QED loops supply the leading anomaly	Established	Schwinger and later QED	Precision perturbation theory

Claim ID	Claim	Status	Established source or author layer	Decisive test
EGF-08	Electroweak dipoles map through Q_{eB} , Q_{eW} , and $C_{e\gamma}$	Established EFT dictionary	Warsaw basis; SMEFT / LEFT literature	Convention-controlled matching
EGF-09	SIT variables may modulate dipole Wilson coefficients through invariant gradients	Proposed hypothesis family	Repaired Version 6 proposal	Differential intervention under fixed response coefficients
EGF-10	The environmental component vanishes at the uniform-gradient reference point	Definition of the proposed family	Equation (28)	Verify the operational fields and baseline

The public genealogy is:

- **24 December 2025, Version 6:** informational holonomy, phase-only entry, $g = 2$ recovery, 4-pi context, precession, and an environmental dipole proposal were joined in one paper.
- **Draft 4 technical correction:** the connection / gradient distinction, SMEFT gauge completion, and experimental boundaries were repaired, but the conditional theorem and environmental mechanism were over-narrowed.
- **Version 2.0 current manuscript:** the valid Version 6 members are restored inside explicit premises; the malformed environmental operator is replaced with Equations (25)-(28); no false historical date or experimental result is added.

13. Scope and limitations

This paper does not derive the existence of a Dirac field from information theory. It does not derive ρ_t , R_{coh} , their dynamics, or the response coefficients κ . It does not calculate SMEFT running, operator mixing, or a precision anomaly bound. It does not claim that the measured electron moment validates SIT.

The paper does accomplish four bounded tasks:

1. it states exactly what must be assumed before phase holonomy selects the minimal Dirac operator;
2. it reproduces the standard factor of two without skipping the magnetic algebra;
3. it preserves the original SIT interpretation while distinguishing it from established spinor and gauge structures;
4. it turns the environmental idea into a gauge-invariant, dimensionally closed, discriminating hypothesis family.

14. Conclusion

The electron g -factor is a useful boundary object between geometry, representation theory, effective field theory, and experiment. A $U(1)$ holonomy does not by itself create the electron or choose the Dirac action. Once a charged Dirac spinor and the phase-only-entry design space are fixed, however, the allowed renormalizable bilinear matter operator is the minimally coupled Dirac form. Its exact nonrelativistic reduction gives the standard tree-level result $g = 2$; the Gordon identity exposes the same coefficient in the form-factor language; QED loops and higher-dimensional operators remain separate layers.

SIT's contribution in this paper is therefore neither the historical discovery of $g = 2$ nor an unsupported anomaly claim. It is a joined structural proposal: informational phase holonomy, a bounded entry rule,

a clear standard-limit derivation, and an environmental EFT bridge that states how a future SIT variable could become a measurable coefficient. The bridge becomes a quantitative prediction only when the scalar fields, response coefficients, scales, and experimental intervention are fixed. That is the exact point at which interpretation must become measurement.

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